



ÉCOLE NATIONALE SUPÉRIEURE DE L'ÉNERGIE,
L'EAU ET L'ENVIRONNEMENT

LAB REPORT

SMART CITIES - 4EUS4HAT

Energy Building modelling

Tran
Lam The Thinh

Muth
Kanhaputheavy

Hosseini
Romina

Montagne
Édouard

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1 Introduction

The link to all .ipynb files can be found in this [GITHUB](#) repo.

First and foremost, we have modeled the room we will focus on. We considered a 30 m^2 apartment with one window and one door. The room is situated in the middle of the building, so that the floor, ceiling and outdoor walls are not in contact with the exterior. The floor is covered with floorboard, the window is made with double glazing glass and the door is made of wood. The walls are 3m tall and are insulated.

The diagram representing our room is shown below :

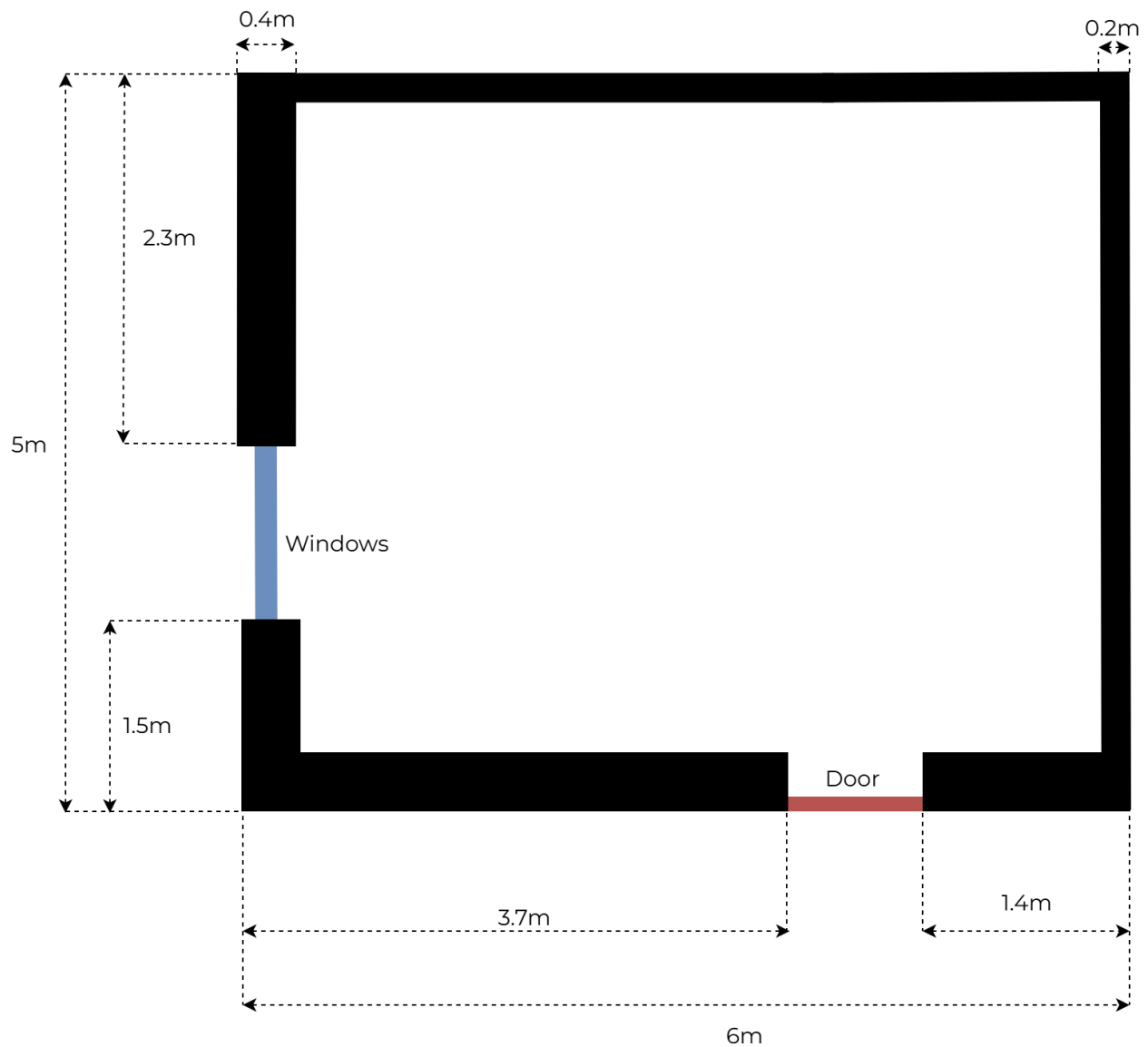


Figure 1: Room plan

1.1 General information

- Location
 - The building is situated in Grenoble, Isere department. Meteo code: 074860.
- Materials
 - Outdoor Wall : Concrete and insulation
 - Window : Glass
 - Door : Wood
- Height $H = 3\text{m}$
- Surfaces
 - $S_g = 1.2 \times 3 = 3.6 \text{ m}^2$ (Surface area of the glass)
 - $S_c = 26.7 \text{ m}^2$ (Surface area of the outdoor wall)
 - $S_b = 2.7 \text{ m}^2$ (Surface area of door)
- Temperature control: HVAC system.

Thermo-Physical properties

	Conductivity λ [W/(m K)]	Density ρ [kg/m ³]	Specific heat c [J/(kg K)]	Width w [m]
Air	-	1.2	1000	-
Wood	0.2	310	1733	0.07
Concrete	1.4	2300	880	0.32
Insulation	0.027	55	1210	0.08
Glass	1.4	2500	1210	0.04

Table 1: Material properties

Radiative properties

$\varepsilon_{w,LW} = 0.85$ Long wave emissivity: outdoor wall surface

$\varepsilon_{g,LW} = 0.90$ Long wave emissivity: glass

$\alpha_{w,SW} = 0.25$ Short wave absorptivity: white smooth surface

$\alpha_{g,SW} = 0.38$ Short wave absorptivity: reflective blue glass

$\tau_{g,SW} = 0.30$ Short wave transmittance: reflective blue glass

1.2 Hypothesis

1. The temperature is uniform on each surface of the wall.
2. Heat transfer with the floor and ceiling is neglected (adiabatic).
3. Heat transfer with the indoor walls between apartments is neglected (adiabatic).
4. Heat transfers are assumed to be linear.
5. The indoor temperature is assumed to be homogeneous.

2 Thermal Circuit

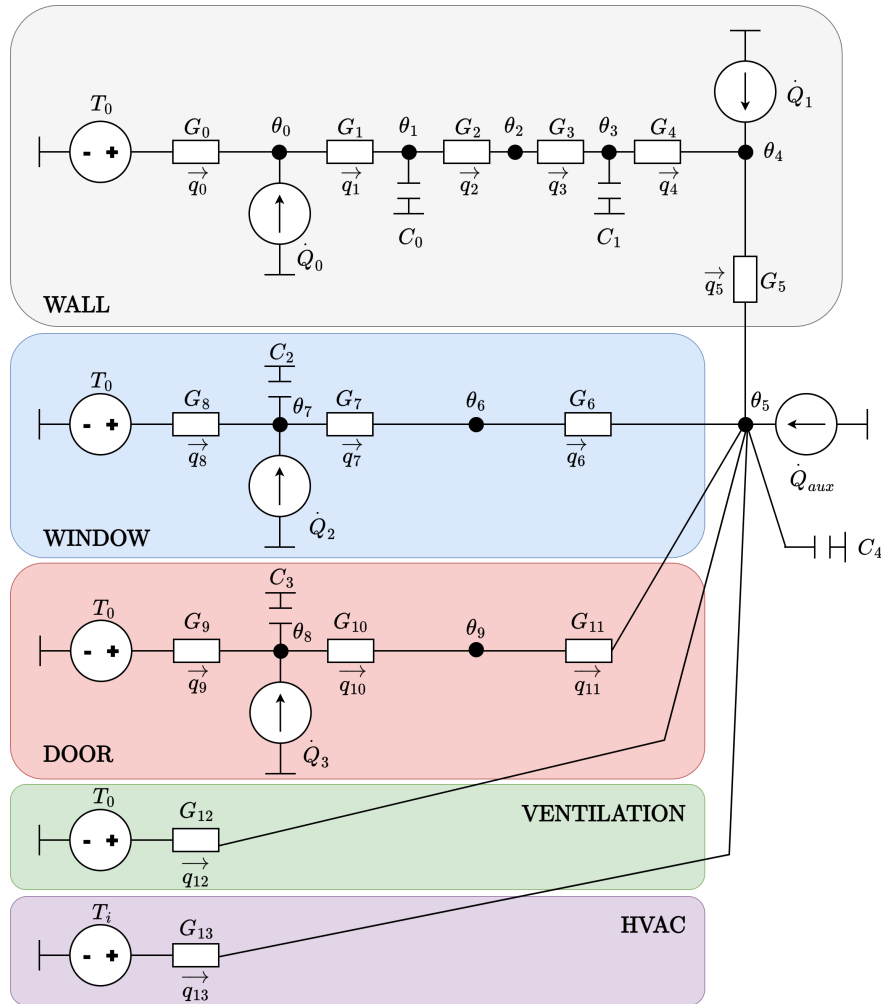


Figure 2: Thermal Circuit

3 Mathematical model

Summary

There are 8 temperature nodes, numbered from θ_0 to θ_7 ; 14 heatflow branches numbered from q_0 to q_{13} with corresponding conductances G_0 to G_{13} ; 1 temperature source T_0 ; 5 capacitances C_0 to C_4 ; 5 heatflow sources $\dot{Q}_0$ to $\dot{Q}_3$ and $\dot{Q}_{aux}$.

3.1 Conductance Matrix G

Wall Section

Conductance	Description	Formula	Calculation	Value (W/K)
G_0	Outdoor Convection	$h_o S_c$	25×26.7	667.50
G_1	Half Concrete Cond.	$2\lambda_c S_c / w_c$	$2 \times 1.4 \times 26.7 / 0.32$	233.63
G_2	Half Concrete Cond.	$2\lambda_c S_c / w_c$	$2 \times 1.4 \times 26.7 / 0.32$	233.63
G_3	Half Insulation Cond.	$2\lambda_i S_c / w_i$	$2 \times 0.027 \times 26.7 / 0.08$	18.02
G_4	Half Insulation Cond.	$2\lambda_i S_c / w_i$	$2 \times 0.027 \times 26.7 / 0.08$	18.02
G_5	Indoor Convection	$h_i S_c$	8×26.7	213.60

Table 2: Thermal conductance values - Wall section

Window Section

Conductance	Description	Formula	Calculation	Value (W/K)
G_8	Outdoor Convection	$h_o S_g$	25×3.6	90.00
G_7	Glass Conduction	$\lambda_g S_g / w_g$	$1.4 \times 3.6 / 0.04$	126.00
G_6	Indoor Convection	$h_i S_g$	8×3.6	28.80

Table 3: Thermal conductance values - window section

Door Section

Conductance	Description	Formula	Calculation	Value (W/K)
G_9	Outdoor Convection	$h_o S_b$	25×2.7	67.50
G_{10}	Half Wood Conduction	$2\lambda_b S_b / w_b$	$2 \times 0.2 \times 2.7 / 0.07$	15.43
G_{11}	Half Wood Conduction	$2\lambda_b S_b / w_b$	$2 \times 0.2 \times 2.7 / 0.07$	15.43

Table 4: Thermal conductance values - Door section

Ventilation and HVAC Section

The conductance matrix G is then given by:

Conductance	Description	Formula	Calculation	Value (W/K)
G_{12}	Advection	$\rho c \dot{V}_a$	$1.2 \times 1000 \times 0.025$	30.00
G_{13}	HVAC Controller	K_p	–	–

Table 5: Thermal conductance values - Ventilation and HVAC Section

$$G = \begin{bmatrix} G_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & G_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & G_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & G_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & G_4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & G_5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & G_6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_7 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_8 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{11} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & G_{13} \end{bmatrix} \quad (1)$$

3.2 Incidence Matrix A

An element $A_{k,j}$ is 1 if flow q_k enters node θ_j , -1 if it leaves, and 0 otherwise. Therefore, based on the thermal circuit from figure 2, the incidence matrix A is given as below.

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (2)$$

3.3 Capacity Matrix C

From the room plan in figure 1, the following capacities are modelled:

- C_0 at node θ_1 : Represents the thermal mass of the concrete layer in the wall.
- C_1 at node θ_3 : Represents the thermal mass of the insulation layer in the wall.
- C_2 at node θ_6 : Represents the thermal mass of the glass window.
- C_3 at node θ_7 : Represents the thermal mass of the wooden door.
- C_4 at node θ_5 : Represents the thermal mass of the indoor air volume.

The C matrix is a diagonal matrix. If the node has a capacity attached to it, the value is assigned. Therefore, the C matrix is achieved as follows.

Capacity	Node	Component	Formula	Calculation	Value (J/K)
C_0	θ_1	Concrete Layer	$\rho_c c_c S_c w_c$	$2300 \cdot 880 \cdot 26.7 \cdot 0.32$	17,293,120
C_1	θ_3	Insulation Layer	$\rho_i c_i S_i w_i$	$55 \cdot 1210 \cdot 26.7 \cdot 0.08$	142,152
C_2	θ_7	Glass Window	$\rho_g c_g S_g w_g$	$2500 \cdot 1210 \cdot 3.6 \cdot 0.04$	435,600
C_3	θ_8	Wooden Door	$\rho_b c_b S_b w_b$	$310 \cdot 1733 \cdot 2.7 \cdot 0.07$	101,507
C_4	θ_5	Indoor Air	$\rho_a c_a V_a$	$1.2 \cdot 1000 \cdot 90$	108,000

Table 6: Thermal capacitance values

$$C = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & C_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & C_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & C_4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & C_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & C_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (3)$$

3.4 Temperature Source Vector b

The temperature source vector assigns a value when the temperature source is attached to the branch of heat flow. Therefore, the vector is achieved as follows.

$$b = [T_0, 0, 0, 0, 0, 0, 0, 0, T_0, T_0, 0, 0, T_0, T_i]^T \quad (4)$$

3.5 Heat Flow Source Vector f

The heat flow source vector assigns a value when the heat flow source is attached to the branch of heat flow. Therefore, the heat flow source vector f is achieved as follows.

Source	Formula	Calculation	Value (W)
$\dot{Q}_0$	$\alpha_{w,SW} S_c E$	$0.25 \cdot 26.7 \cdot 200$	667.50
$\dot{Q}_1$	$\tau_{g,SW} \alpha_{w,SW} S_g E$	$0.30 \cdot 0.25 \cdot 3.6 \cdot 200$	54.00
$\dot{Q}_2$	$\alpha_{g,SW} S_g E$	$0.38 \cdot 3.6 \cdot 200$	273.60
$\dot{Q}_3$	$\alpha_{w,SW} S_b E$	$0.25 \cdot 2.7 \cdot 200$	135.00
$\dot{Q}_{aux}$	—	Assumed (e.g., 0 or 1000)	0.00

Table 7: Heat source values

$$f = [\dot{Q}_0, 0, 0, 0, \dot{Q}_1, \dot{Q}_{aux}, 0, \dot{Q}_2, \dot{Q}_3, 0]^T \quad (5)$$

4 Thermal load and yearly consumption

In steady-state (all time derivatives vanish, $\dot{\theta} = 0$), the DAE system reduces to a purely algebraic system. For a **perfect HVAC controller** ($K_p \rightarrow \infty$), the indoor temperature θ_5 is forced to the setpoint T_i . The HVAC load then equals the total net heat exchange with the exterior.

The **overall thermal conductance** between the outdoor air and the indoor air is computed by combining, in series, the conductances along each heat-transfer path:

$$UA_{\text{wall}} = \frac{1}{\frac{1}{G_0} + \frac{1}{G_1} + \frac{1}{G_2} + \frac{1}{G_3} + \frac{1}{G_4} + \frac{1}{G_5}} = \frac{1}{\frac{1}{667.50} + \frac{2}{233.63} + \frac{2}{18.02} + \frac{1}{213.60}} = \mathbf{7.95 \text{ W/K}} \quad (6)$$

$$UA_{\text{win}} = \frac{1}{\frac{1}{G_8} + \frac{1}{G_7} + \frac{1}{G_6}} = \frac{1}{\frac{1}{90.00} + \frac{1}{126.00} + \frac{1}{28.80}} = \mathbf{18.60 \text{ W/K}} \quad (7)$$

$$UA_{\text{door}} = \frac{1}{\frac{1}{G_9} + \frac{1}{G_{10}} + \frac{1}{G_{11}}} = \frac{1}{\frac{1}{67.50} + \frac{2}{15.43}} = \mathbf{6.92 \text{ W/K}} \quad (8)$$

$$H_{\text{total}} = UA_{\text{wall}} + UA_{\text{win}} + UA_{\text{door}} + G_{12} = 7.95 + 18.60 + 6.92 + 30.00 = \mathbf{63.47 \text{ W/K}} \quad (9)$$

Table 8: Overall heat transfer coefficients

Element	Series path	UA [W/K]	U [W/(m ² · K)]	Surface [m ²]
Outdoor wall	$G_0 \rightarrow G_1 \rightarrow G_2 \rightarrow G_3 \rightarrow G_4 \rightarrow G_5$	7.95	0.298	26.7
Window	$G_8 \rightarrow G_7 \rightarrow G_6$	18.60	5.17	3.6
Door	$G_9 \rightarrow G_{10} \rightarrow G_{11}$	6.92	2.56	2.7
Ventilation	G_{12} (advection)	30.00	—	—
Total H_{total}	All paths	63.47	—	—

4.1 Thermal load

4.1.1 Most unfavorable condition

The most unfavorable conditions are defined below, together with the resulting HVAC load.

Maximum Heating Load (Winter)

- Lowest outdoor temperature: $T_o = -10$ °C (Grenoble design winter temperature, RT2012/RE2020)
- Indoor setpoint: $T_{i,sp} = 20$ °C
- Cloudy sky: no solar radiation ($E = 0$ W/m²)
- No internal heat sources ($\dot{Q}_{\text{aux}} = 0$ W) — worst-case night scenario
- Ventilation active: ACH = 1 h⁻¹ ($\dot{V}_a = 0.025$ m³/s, $G_{12} = 30$ W/K)

$$\Delta T_{\text{win}} = T_{i,sp} - T_o = 20 - (-10) = 30 \text{ K} \quad (10)$$

Table 9: Heating load breakdown — most unfavorable winter condition

Heat Loss Path	Formula	Value [W]
Outdoor wall	$UA_{\text{wall}} \times \Delta T = 7.95 \times 30$	238.5
Window	$UA_{\text{win}} \times \Delta T = 18.60 \times 30$	558.0
Door	$UA_{\text{door}} \times \Delta T = 6.92 \times 30$	207.6
Ventilation	$G_{12} \times \Delta T = 30.00 \times 30$	900.0
HVAC heating load $\dot{Q}_{\text{heat}}$	$H_{\text{total}} \times \Delta T = 63.47 \times 30$	1904.1

$$\dot{Q}_{\text{heat}} = H_{\text{total}} \times \Delta T_{\text{win}} = 63.47 \times 30 \approx \mathbf{1904} \text{ W} \approx 1.90 \text{ kW} \quad (11)$$

Specific heating power: $\dot{Q}_{\text{heat}}/A_{\text{floor}} = 1904/30 \approx 63.5$ W/m².

Maximum Cooling Load (Summer)

- Highest outdoor temperature: $T_o = 35$ °C (Grenoble peak summer, RT2012/RE2020)
- Indoor setpoint: $T_{i,sp} = 26$ °C
- Peak solar irradiance: $E = 500$ W/m² on south-facing surfaces
- No internal heat sources ($\dot{Q}_{aux} = 0$ W)

$$\Delta T_{sum} = T_o - T_{i,sp} = 35 - 26 = 9 \text{ K} \quad (12)$$

The total cooling load includes the envelope transmission, solar radiation transmitted through the window, and solar gains conducted through the opaque surfaces:

$$\dot{Q}_{env} = H_{total} \times \Delta T_{sum} = 63.47 \times 9 = 571.2 \text{ W} \quad (13)$$

$$\dot{\Phi}_{win} = \tau_{g,SW} \cdot S_g \cdot E = 0.30 \times 3.6 \times 500 = 540.0 \text{ W} \quad (14)$$

$$\dot{\Phi}_{wall} = \alpha_{w,SW} \cdot S_c \cdot E \cdot \frac{U A_{wall}}{G_0} = 0.25 \times 26.7 \times 500 \times \frac{7.95}{667.50} = 39.8 \text{ W} \quad (15)$$

$$\dot{\Phi}_{door} = \alpha_{w,SW} \cdot S_b \cdot E \cdot \frac{U A_{door}}{G_9} = 0.25 \times 2.7 \times 500 \times \frac{6.92}{67.50} = 34.6 \text{ W} \quad (16)$$

Table 10: Cooling load breakdown — most unfavorable summer condition

Cooling Load Component	Formula	Value [W]
Envelope transmission	$H_{total} \times \Delta T = 63.47 \times 9$	571.2
Solar gain through window	$\tau_{g,SW} \cdot S_g \cdot E = 0.30 \times 3.6 \times 500$	540.0
Solar gain via wall (cond.)	$\alpha_{w,SW} \cdot S_c \cdot E \cdot (U A_{wall}/G_0)$	39.8
Solar gain via door (cond.)	$\alpha_{w,SW} \cdot S_b \cdot E \cdot (U A_{door}/G_9)$	34.6
HVAC cooling load $\dot{Q}_{cool}$	Sum of above	1185.6

$$\boxed{\dot{Q}_{cool} = 1185.6 \text{ W} \approx 1.19 \text{ kW}} \quad (17)$$

Table 11: Thermal load summary — most unfavorable conditions

Condition	T_o [°C]	$T_{i,sp}$ [°C]	ΔT [K]	E [W/m ²]	HVAC Load [W]
Winter (heating)	−10	20	30	0	1904
Summer (cooling)	35	26	9	500	1186

Conclusion The **winter heating condition is the most unfavorable**, producing the higher HVAC demand. The summer cooling load is dominated by solar gains through the south-facing window ($\dot{\Phi}_{win} = 540$ W, representing 45.6% of the total cooling load).

4.2 Yearly energy consumption

Annual energy consumption is estimated using the **degree-day method**, which integrates the thermal balance over the year using Heating Degree Days (HDD) and Cooling Degree Days (CDD).

Table 12: Climate data for Grenoble

Parameter	Value	Unit	Notes
Heating Degree Days (base 20°C)	3 260	°C·day/year	Continental alpine climate
Cooling Degree Days (base 26°C)	150	°C·day/year	Mild summers in alpine valley
Solar irradiation — south vertical	1 200	kWh/(m ² ·year)	Annual total, south façade
Heating season duration	≈ 212	days/year	October – April
Cooling season duration	≈ 90	days/year	June – August

Climate Data — Grenoble (Météo France, code 074860)

Heating Energy The gross annual heating demand is computed from the degree-day formula:

$$E_{\text{heat,gross}} = H_{\text{total}} \times \text{HDD} \times 24 \frac{\text{h}}{\text{day}} = 63.47 \times 3260 \times 24 = 4,967,443 \text{ Wh/year} \approx \mathbf{4\,967 \text{ kWh/year}} \quad (18)$$

Useful solar gains through the south-facing window reduce the net demand. A solar utilization factor $\eta_{\text{sol}} = 0.50$ accounts for the fact that gains are not always available when heating is needed:

$$E_{\text{solar}} = \tau_{g,SW} \cdot S_g \cdot H_{\text{south}} \cdot \eta_{\text{sol}} = 0.30 \times 3.6 \times 1200 \times 0.50 = \mathbf{648 \text{ kWh/year}} \quad (19)$$

Net heating demand and delivered energy, assuming HVAC system efficiency $\eta_{\text{HVAC}} = 0.85$:

$$E_{\text{heat,net}} = E_{\text{heat,gross}} - E_{\text{solar}} = 4967 - 648 = \mathbf{4\,319 \text{ kWh/year}} \quad (20)$$

$$E_{\text{heat,delivered}} = \frac{E_{\text{heat,net}}}{\eta_{\text{HVAC}}} = \frac{4319}{0.85} = \mathbf{5\,081 \text{ kWh/year}} \quad (21)$$

Cooling Energy Cooling energy is estimated from the CDD method plus solar gains through the window during summer, using a coefficient of performance $\text{COP} = 3.5$:

$$E_{\text{cool,env}} = \frac{H_{\text{total}} \times \text{CDD} \times 24}{\text{COP} \times 1000} = \frac{63.47 \times 150 \times 24}{3.5 \times 1000} = \mathbf{65.3 \text{ kWh/year}} \quad (22)$$

$$E_{\text{cool,solar}} = \frac{\tau_{g,SW} \cdot S_g \cdot H_{\text{summer}}}{\text{COP}} = \frac{0.30 \times 3.6 \times 300}{3.5} = \mathbf{92.6 \text{ kWh/year}} \quad (23)$$

$$E_{\text{cool,delivered}} = E_{\text{cool,env}} + E_{\text{cool,solar}} = \mathbf{157.9 \text{ kWh/year}} \quad (24)$$

Table 13: Annual energy breakdown

Energy Component	Value [kWh/year]	Share	Comment
Gross heating demand	4 967	—	Degree-day method, $H = 63.47$ W/K
Solar gains offset (heating)	−648	—	South window, $\eta_{\text{sol}} = 0.50$
Net heating demand	4 319	—	After solar offset
Heating energy delivered	5 081	97%	$\eta_{\text{HVAC}} = 0.85$
Cooling energy delivered	158	3%	$\text{COP} = 3.5$
Total E_{total}	5 239	100%	
Floor area A_{floor}	30 m ²	—	
Specific consumption e	175 kWh/(m²·year)	—	RT2012 ref: 50–80 kWh/(m ² ·yr)

Annual Energy Summary The specific energy consumption of **175 kWh/(m² · year)** is dominated by heating (97% of total), consistent with Grenoble’s continental alpine climate. The main driver is ventilation loss: $G_{12} = 30$ W/K contributes 47% of H_{total} . Installing mechanical heat recovery ventilation (MVHR) with 80% efficiency would reduce the effective ventilation conductance to approximately 6 W/K, saving roughly 1 500 kWh/year in heating energy alone.

5 Dynamic modeling

The thermal circuit derived in the previous report is given as in Figure 2.

The derived matrices and vectors are:

$$A = \begin{array}{c|cccccccccc} & \theta_0 & \theta_1 & \theta_2 & \theta_3 & \theta_4 & \theta_5 & \theta_6 & \theta_7 & \theta_8 & \theta_9 \\ \hline q_0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ q_1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ q_2 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ q_3 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ q_4 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ q_5 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ q_6 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ q_7 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ q_8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ q_9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ q_{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \\ q_{11} & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 \\ q_{12} & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ q_{13} & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array} \quad (25)$$

$$G = \text{diag}(667.5, 233.63, 233.63, 18.02, 18.02, 213.6, 28.8, 126, 90, 67.5, 15.43, 15.43, 30, 0) \quad (26)$$

$$C = \text{diag}(0, 17,293,120, 0, 142,152, 0, 108,000, 0, 435,600, 101,507, 0) \quad (27)$$

$$b = (T_o \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ T_o \ T_o \ 0 \ 0 \ T_o \ T_i)^T \quad (28)$$

$$f = (667.5 \ 0 \ 0 \ 0 \ 54 \ 0 \ 0 \ 273.6 \ 135 \ 0)^T \quad (29)$$

$$y = (0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0)^T \quad (30)$$

These matrices are then input into the .csv file as follows, to utilise the sample code from [1]:

```

1  A,00,01,02,03,04,05,06,07,08,09,G,b
2  q0,1,,,,,,,,,667.5,To
3  q1,-1,1,,,,,,,,,233.63,
4  q2,, -1,1,,,,,,,,,233.63,
5  q3,,, -1,1,,,,,,,,,18.02,
6  q4,,,, -1,1,,,,,,,,,18.02,
7  q5,,,,, -1,1,,,,,213.6,
8  q6,,,,,, 1,-1,,,28.8,
9  q7,,,,,, 1,-1,,126,
10 q8,,,,,, 1,,90,To
11 q9,,,,,, 1,,67.5,To
12 q10,,,,,, -1,1,15.43,
13 q11,,,,, 1,,, -1,15.43,
14 q12,,,,, 1,,,,,30,To
15 q13,,,,, 1,,,,,Ti
16 C,,17293120,,142152,,108000,,435600,101507,,
17 f,667.5,,,54,,,273.6,135,
18 y,,,,,1,,,,,
19
```

Figure 3: Matrices input in .csv

6 Steady-state analysis

Consider the following boundary conditions as a falsification test:

- Outdoor temperature: $T_o = 10^\circ\text{C}$ (applied to branches q_0, q_8, q_9, q_{12}),
- Indoor setpoint temperature: $T_i = 20^\circ\text{C}$ (on branch q_{13}),

- All flow-rate sources: $f_{ss} = 0$.

6.1 Steady-state from differential algebraic equations

The dynamic behavior of the thermal circuit is described by the following system of differential-algebraic equations [1]:

$$\begin{cases} C\dot{\theta} = -(A^TGA)\theta + A^TGb + f \\ q = G(-A\theta + b) \end{cases} \quad (31)$$

where:

- $\theta \in R^n$ is the vector of node temperatures ($^{\circ}\text{C}$),
- $q \in R^m$ is the vector of branch heat fluxes (W),
- $A \in R^{m \times n}$ is the **incidence matrix** of the thermal network, with entries +1, -1, or 0 depending on branch orientation with respect to nodes,
- $G = \text{diag}(G_0, G_1, \dots, G_{m-1}) \in R^{m \times m}$ is the diagonal matrix of **branch conductances** (W/K),
- $C = \text{diag}(C_0, C_1, \dots, C_{n-1}) \in R^{n \times n}$ is the diagonal matrix of **nodal thermal capacitances** (J/K),
- $b \in R^m$ is the vector of **temperature sources** on branches ($^{\circ}\text{C}$),
- $f \in R^n$ is the vector of **flow-rate sources** (heat injected) at nodes (W),
- $\dot{\theta} = d\theta/dt$ is the time derivative of node temperatures.

```
bss = np.zeros(14)
bss[[0, 8, 9, 12]] = Tout      # outdoor temperature
bss[[13]] = Tin                # indoor set-point temperature

fss = np.zeros(10)

A = TC['A']
G = TC['G']
diag_G = pd.DataFrame(np.diag(G), index=G.index, columns=G.index)

theta_ss = np.linalg.inv(A.T @ diag_G @ A) @ (A.T @ diag_G @ bss + fss)
print(f'theta_ss = {np.around(theta_ss, 2)} °C')
```

Figure 4: Steady-state node temperatures when the flow sources are zero code

```
Matrices and vectors for thermal circuit from TC_ss.csv
θss = [10. 10. 10. 10. 10. 10. 10. 10. 10. 10.] °C
```

Figure 5: Steady-state node temperatures when the flow sources are zero

```
bss_Q = np.zeros(14)

fss_Q = np.zeros(10)
fss_Q[0] = 667.5
fss_Q[4] = 54.0
fss_Q[7] = 273.6
fss_Q[8] = 135.0

θssQ = np.linalg.inv(A.T @ diag_G @ A) @ (A.T @ diag_G @ bss_Q + fss_Q)
print(f'θssQ = {np.around(θssQ, 2)} °C')
```

Figure 6: Steady-state node temperatures when the flow sources are not zero code

```
θssQ = [1.02 1.06 1.1 1.68 2.26 2.05 2.69 2.84 2.01 2.03] °C
```

Figure 7: Steady-state node temperatures when the flow sources are not zero

6.2 Steady-state from state-space representation

Since not all nodes necessarily have a capacitance associated with them, the DAE system can be reduced to a state-space representation. Only nodes with non-zero capacitances appear as state variables θ_i . The state-space form is:

$$\begin{cases} \dot{\theta}_s = A_s \theta_s + B_s u \\ y = C_s \theta_s + D_s u \end{cases} \quad (32)$$

where:

- $\theta_s \in R^{n_s}$ is the **state vector** of temperatures at nodes with non-zero thermal capacitance (the “thermal masses”) (°C),
- $u \in R^p$ is the **input vector**, which stacks the non-zero temperature sources b_T and flow-rate sources f_Q :

$$u = \begin{bmatrix} b_T \\ f_Q \end{bmatrix} \quad (33)$$

- $y \in R^q$ is the **output vector** (e.g., the indoor air temperature),
- $A_s \in R^{n_s \times n_s}$ is the **state matrix**,
- $B_s \in R^{n_s \times p}$ is the **input matrix**,
- $C_s \in R^{q \times n_s}$ is the **output matrix**,
- $D_s \in R^{q \times p}$ is the **feedthrough (direct) matrix**.

```

As = (5, 5)
As =
01      03      05      07      08
01 -95909.450518 -446.804726 -78.197970 -65.174593 -7.538747
03 -54354.829583 -5068.738597 -887.110288 -739.367689 -85.522684
05 -12521.174807 -1167.634275 -1701.454347 -1418.087903 -164.030273
07 -2587.401436 -241.282359 -351.592042 -4132.889013 -33.895555
08 -1284.329770 -119.767313 -174.522639 -145.456999 -1366.382950
Bs =
01      03      05      07      08
01 0.00001 0.000000 0.000000 0.000000 0.0 1.499228e-08 0.000000e+00 0.000000 0.000000
03 0.00000 0.000000 0.000000 0.000000 0.0 0.000000e+00 5.473004e-07 0.000000 0.000000
05 0.00000 0.000000 0.000000 0.000278 0.0 0.000000e+00 8.538890e-06 0.000000 0.000000
07 0.00000 0.000207 0.000000 0.000000 0.0 0.000000e+00 0.000000e+00 0.000002 0.000000
08 0.00000 0.000000 0.000665 0.000000 0.0 0.000000e+00 0.000000e+00 0.000000 0.000001
Bs = (5, 9)
Cs =
01      03      05      07      08
05 0.0 0.0 1.0 0.0 0.0
Ds =
01      03      05      07      08
05 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0

```

Figure 8: State matrices

```

bT = np.array([Tout, Tout, Tout, Tout, Tin]) # [To, To, To, To, Ti]
fQ = np.array([0, 0, 0, 0]) # constant flow sources at nominal values
uss = np.hstack([bT, fQ])
print(f'uss = {uss}')

inv_As = pd.DataFrame(np.linalg.inv(As),
| | | | | | columns=As.index, index=As.index)

yss = (-Cs @ inv_As @ Bs + Ds) @ uss

yss = float(yss.values[0])
print(f'yss = {yss:.2f} °C')

print(f'Error between DAE and state-space: {abs(theta[5] - yss):.2e} °C')

```

Figure 9: Steady state input and output vector when all flow sources are zero code

```
uss = [10 10 10 10 20 0 0 0 0]
yss = 10.00 °C
Error between DAE and state-space: 3.55e-15 °C
```

Figure 10: Steady state input and output vector when all flow sources are zero

```
bT_Q = np.array([0, 0, 0, 0, 0])
fQ_Q = np.array([667.5, 54.0, 273.6, 135.0])
uss_Q = np.hstack([bT_Q, fQ_Q])

inv_As = pd.DataFrame(np.linalg.inv(As),
                      columns=As.index, index=As.index)
yssQ = (-Cs @ inv_As @ Bs + Ds) @ uss_Q

yssQ = float(yssQ.values[0])
print(f'uss = {uss_Q}')
print(f'yssQ = {yssQ:.2f} °C')

print(f'Error between DAE and state-space: {abs(0ssQ[5] - yssQ):.2e} °C')
```

Figure 11: Steady state input and output vector when all flow sources are not zero code

```
uss = [ 0. 0. 0. 0. 0. 667.5 54. 273.6 135. ]
yssQ = 2.05 °C
Error between DAE and state-space: 1.33e-15 °C
```

Figure 12: Steady state input and output vector when all flow sources are not zero

7 Eigenvalues analysis

```
λ = [-8.31053087e-04 -6.86907311e-04 -1.03961171e-05 -1.86199345e-04
      -2.52558708e-04]
```

Figure 13: System eigenvalues

7.1 Time step

The maximum stable time step for the Euler explicit integration method is determined by the most negative eigenvalue $\lambda_{\min}$ of the state matrix A_s :

$$\Delta t_{\max} = \frac{-2}{\lambda_{\min}} \quad (34)$$

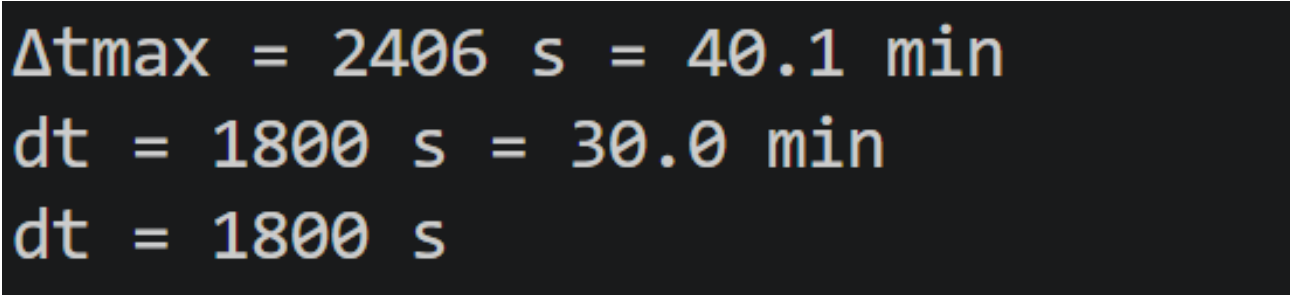
For this system, the computed value is:

$$\Delta t_{\max} = 2406 \text{ s} = 40.1 \text{ min} \quad (35)$$

To ensure the time step aligns with the weather data resolution, Δt is rounded down to the nearest value that divides evenly into the data period:

$$\Delta t = 1800 \text{ s} = 30.0 \text{ min} \quad (36)$$

This value is safely below $\Delta t_{\max}$, guaranteeing numerical stability. Using a time step equal to the weather data resolution also avoids unnecessary interpolation error in the input signals.



```
dtmax = 2406 s = 40.1 min
dt = 1800 s = 30.0 min
dt = 1800 s
```

Figure 14: Time step of the system

7.2 Settling time and duration

The settling time is the time required for the system to move on from the transient state to reach a quasi-steady response driven by the weather inputs. It is estimated as four times the largest time constant $T_{\max}$ of the system:

$$t_{\text{settle}} \approx 4 \cdot T_{\max} = 384,759 \text{ s} = 106.9 \text{ h} \quad (37)$$

The simulation duration is then rounded up to the nearest multiple of Δt that is greater than or equal to t_{settle} :

$$t_{\text{duration}} = 385,200 \text{ s} = 107.0 \text{ h} \quad (38)$$

```
t_settle = 384759 s = 106.9 h
duration = 385200 s = 107.0 h
```

Figure 15: Settling time and duration of the system

8 Step response to outdoor temperature

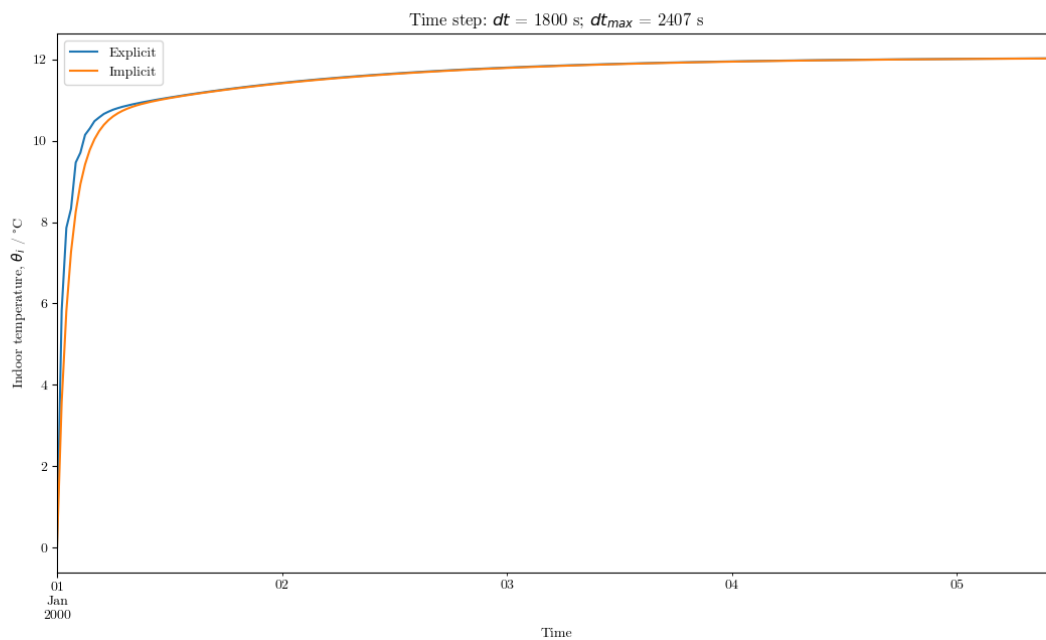


Figure 16: Step response to outdoor temperature by using Euler implicit and explicit integration.

```
Steady-state indoor temperature obtained with:
- DAE model: 10.0000 ^\circ C
- state-space model: 10.0000 ^\circ C
- steady-state response to step input: 12.0253 ^\circ C
```

Figure 17: Step response

9 Perfect controller simulation with weather data

The weather data was taken from the website climate.onebuilding.org/WMO_Region_6_Europe with the file of FRA_AR_Grenoble.Alpes.Isere.AP.074860_TMYx.2011-2025.epw.

The simulated data is from 2025. We will simulate the system response with 2 scenarios, namely the summer and winter scenarios. Firstly, we simulate the system with no controller, then with a perfect controller for comparison.

For the summer scenario, we simulated from 1st of July to the 31th of July, as this is the hottest month in Grenoble on average.

For the winter scenario, we simulated from 1st of January to the 31th of January, as this is the coldest month in Grenoble on average.

9.1 Free-running, no controller

In free-running mode, no heating or cooling is applied. The indoor temperature evolves freely under outdoor weather, solar irradiance, and internal heat gains only. The large time step ($\Delta t = 1800\text{s}$) is numerically stable because the fast controller dynamics are absent.

9.1.1 Summer simulation - July

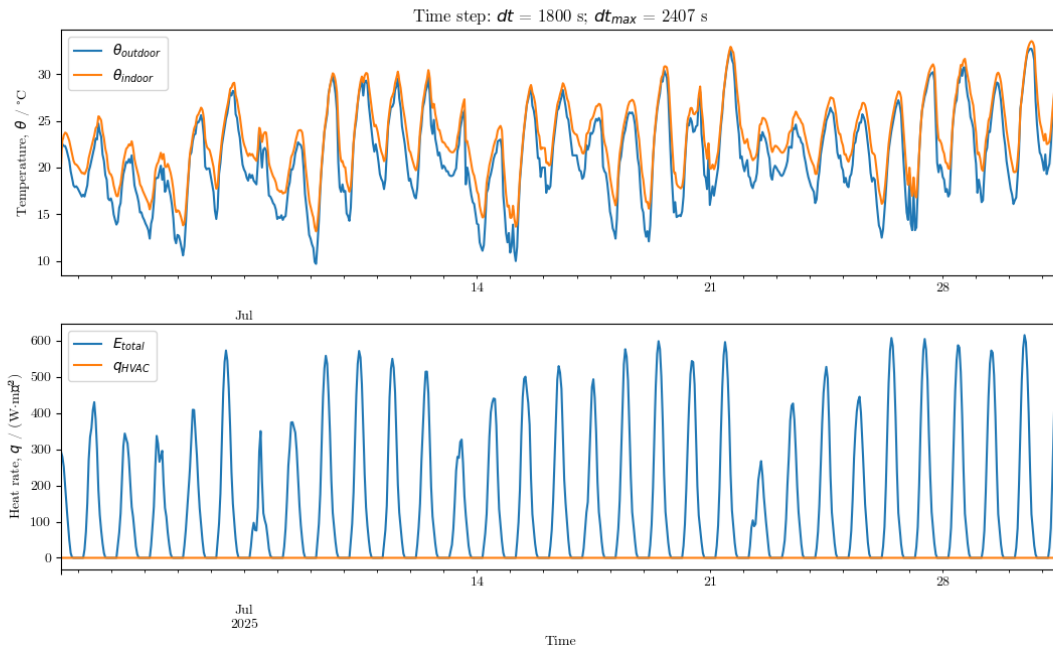


Figure 18: July free-running dynamic response

```
Time step: = 1800 s = 30.0 min
Mean outdoor temperature: 21.1 °C
Min. indoor temperature: 13.2 °C
Max. indoor temperature: 33.5 °C
Max. load: -0.0 W at 2025-07-01 12:00:00+01:00
Energy consumption for heating: 0.0 kWh
Energy consumption for cooling: -0.0 kWh
```

Figure 19: July free-running dynamic response summary

Without a controller, the indoor temperature floats freely between 13.2 °C and 33.5 °C over the month of July. The upper plot in Figure 18 shows the indoor temperature (orange) closely tracking the outdoor temperature (blue), with a slight delay due to the thermal mass of the walls. The indoor temperature exceeds 33 °C on multiple days, well above the standard comfort band of 20–26 °C.

The lower plot in Figure 19 confirms that $q_{HVAC} = 0$ throughout the simulation. The total solar irradiance E_{total} shows strong daily peaks reaching up to $\sim 600 \text{ W/m}^2$, which are the primary driver of daytime indoor overheating. No energy is consumed for heating or cooling. This result demonstrates that the building cannot self-regulate its temperature in summer: active cooling would be necessary to maintain comfort.

9.1.2 Winter simulation - January

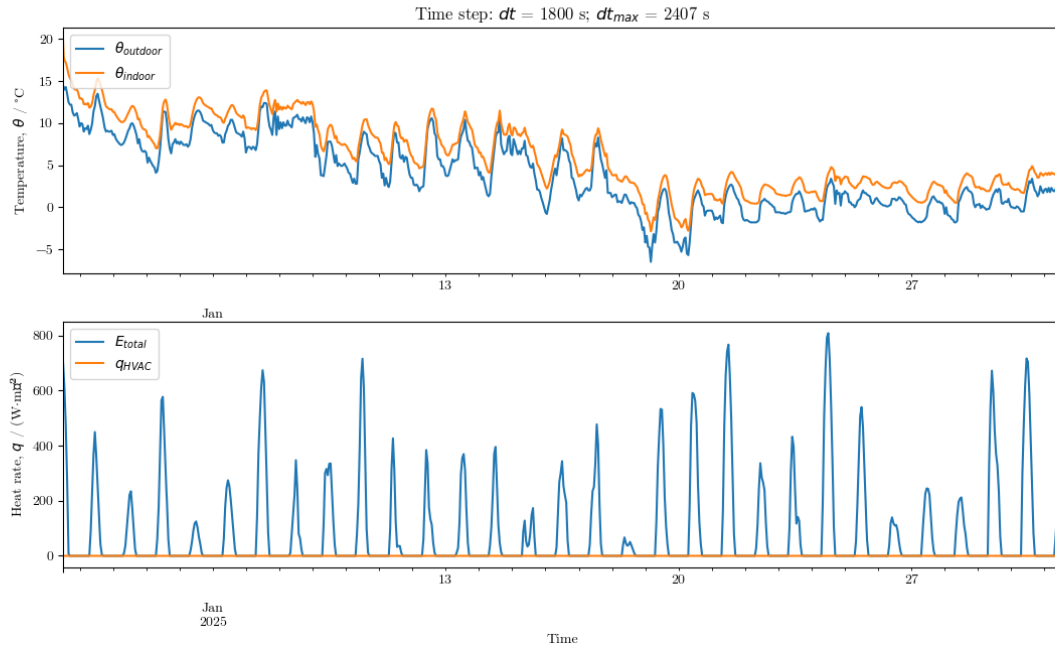


Figure 20: January free-running dynamic response

```

Time step: = 1800 s = 30.0 min
Mean outdoor temperature: 3.9 °C
Min. indoor temperature: -2.9 °C
Max. indoor temperature: 20.0 °C
Max. load: 0.0 W at 2025-01-01 12:00:00+01:00
Energy consumption for heating: 0.0 kWh
Energy consumption for cooling: -0.0 kWh

```

Figure 21: January free-running dynamic response summary

In January, the free-running indoor temperature ranges from -2.9 °C to 20.0 °C . The upper plot in Figure 20 shows outdoor temperatures declining steadily through the month, reaching a minimum of approximately -5.6 °C around January 19–20. The indoor temperature follows closely, dropping below 0 °C during the coldest period.

Brief indoor temperature peaks near 20 °C occur at the start of the month, when solar irradiance provides passive warming during milder days. However, as the month progresses and outdoor temperature falls, solar gains are insufficient to maintain liveable conditions. The q_{HVAC} signal is zero throughout, and no energy is consumed. This confirms that heating is essential in winter: without a controller, the building becomes uninhabitable at sub-zero indoor temperatures.

9.2 With controller

The controller output q_{HVAC} represents the instantaneous heating (positive) or cooling (negative) power applied to the indoor air node. A daily setpoint schedule is applied: $\theta_{sp} = 20$ °C during occupied hours and $\theta_{sp} = 16$ °C at night (setback). The controller gain used was $1e4$. It is suspected that the higher the gain, the lower the time step, hence the lower response time. However, the lower the response time, the higher the power it has to use, and vice versa.

9.2.1 Summer simulation - July

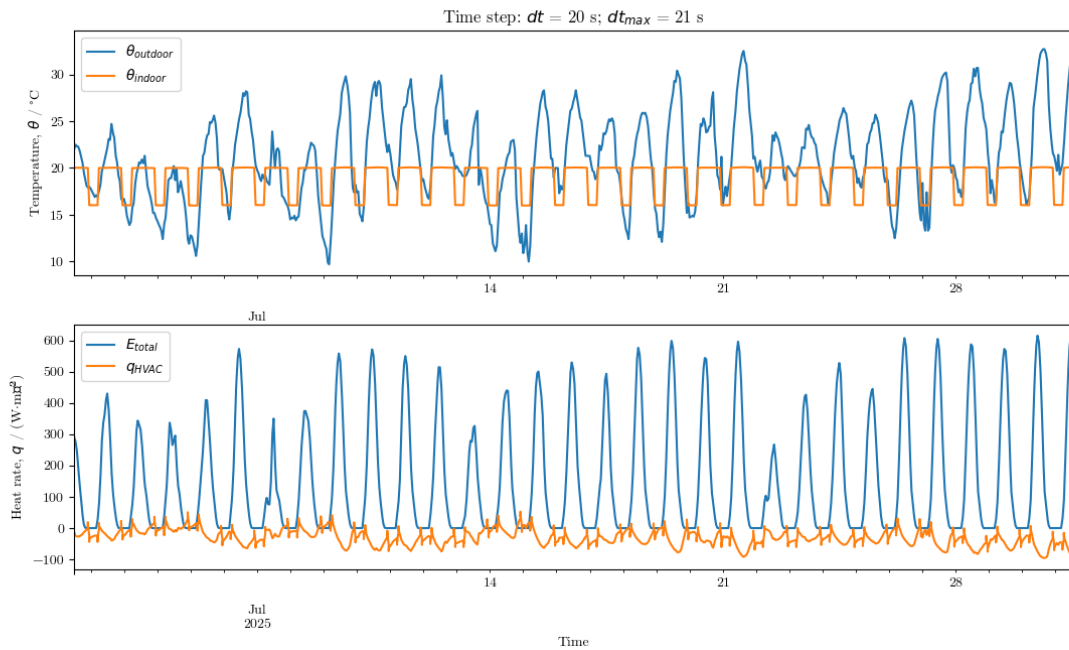


Figure 22: July controlled dynamic response

```

Time step: = 20 s
Mean outdoor temperature: 21.1 °C
Min. indoor temperature: 16.0 °C
Max. indoor temperature: 20.1 °C
Max. load: 52.7 W at 2025-07-14 22:00:20+01:00
Energy consumption for heating: 0.8 kWh
Energy consumption for cooling: 22.5 kWh

```

Figure 23: July controlled dynamic response summary

With the controller active, the indoor temperature in July is maintained tightly between 16.0 °C and 20.1 °C, as shown in the upper plot of Figure 22. The orange line is nearly flat at 20 °C

during occupied hours and drops to the 16 °C setback at night. Comfort is maintained throughout the month despite outdoor temperatures and solar gains that would otherwise drive the indoor temperature above 33 °C.

The lower plot shows q_{HVAC} taking negative values (cooling) during daytime when solar irradiance is high, reaching approximately -100 W/m^2 . Small positive heating pulses appear in the early morning when the indoor temperature rises from the night setback to the daytime setpoint. Total energy consumption over July is 22.5 kWh for cooling and 0.8 kWh for heating, confirming that the summer period is cooling-dominated. The peak load of only 52.7 W occurs at night on July 14, reflecting good passive performance when solar gains are absent.

9.2.2 Winter simulation - January

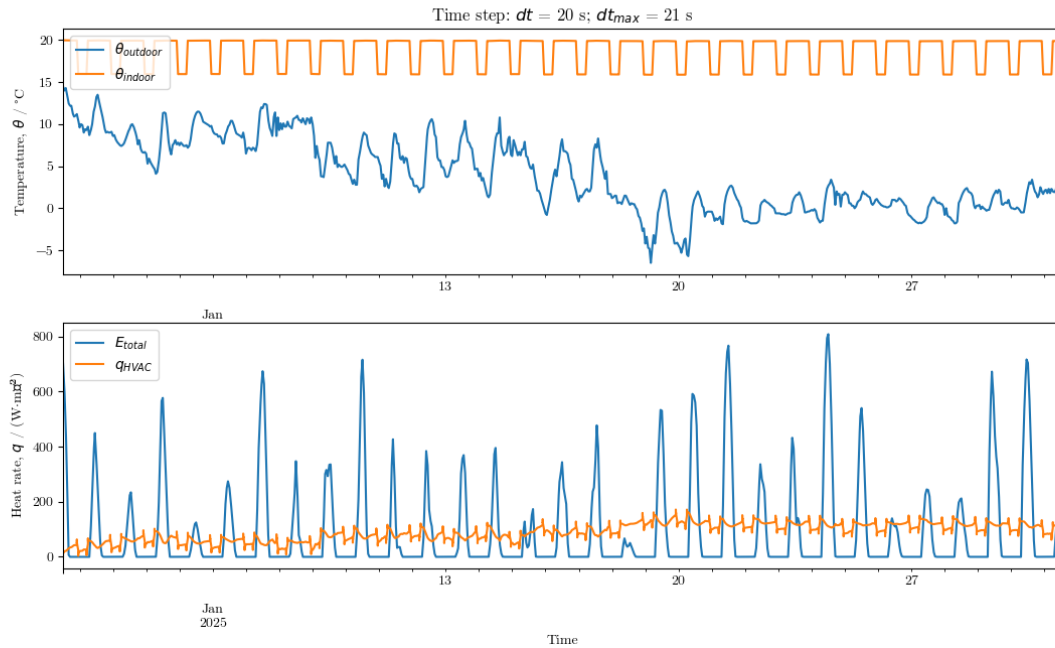


Figure 24: January controlled dynamic response

```
Time step: = 20 s
Mean outdoor temperature: 3.9 °C
Min. indoor temperature: 15.9 °C
Max. indoor temperature: 20.0 °C
Max. load: 171.9 W at 2025-01-19 22:00:20+01:00
Energy consumption for heating: 64.3 kWh
Energy consumption for cooling: -0.0 kWh
```

Figure 25: January controlled dynamic response summary

In January, the controller maintains the indoor temperature between 15.9 °C and 20.0 °C, as shown in Figure ???. The upper plot shows a clear day-night pattern: the indoor temperature (orange) alternates between the 20 °C occupied setpoint and the 16 °C night setback, regardless of how cold it is outside. Outdoor temperatures fall as low as −5.6 °C, but the indoor environment remains comfortable at all times.

The lower plot shows q_{HVAC} is entirely positive (heating only) throughout January. The heating load increases noticeably during the coldest period around January 19–21, where higher sustained heating power is required to compensate for the large indoor-outdoor temperature difference. The total heating energy for January is 64.3 kWh, with a peak load of 171.9 W occurring at night on January 19 — the coldest point of the simulation. No cooling is required (0.0 kWh), confirming that January is heating-dominated. The winter heating demand (64.3 kWh) is approximately 3 times greater than the summer cooling demand (22.5 kWh), highlighting the severity of thermal losses through the building envelope when outdoor temperatures are well below the setpoint.

10 Nonlinear controller

This section investigates two nonlinear control strategies that impose physically meaningful constraints on the HVAC heat rate q_{HVAC} , making them better suited to real building systems. The first strategy, the Proportional Nonlinear Controller (PNC), computes the HVAC command proportionally to the instantaneous temperature error. The second, the Proportional-Integral with saturation Nonlinear Controller (PINC), has an integral term, an anti-windup mechanism, and a saturation law, allowing the controller to eliminate steady-state offset while preventing unbounded accumulation of the integral state and keep the controller output under physical limit.

Both controllers are evaluated across 5 representative scenarios, namely winter heating, summer cooling, combined heating and cooling, solar protection, and passive cooling. The results are analysed in terms of thermal comfort (indoor temperature tracking), energy consumption, and HVAC power smoothness.

These graphs establish the reference: without any HVAC, the indoor temperature reaches 33.5 °C in summer and drops to −2.9 °C in winter — both far outside the comfort band of. The heat rate panels confirm that $q_{HVAC} = 0$ throughout, meaning all thermal dynamics are driven purely by the building envelope and solar gains. This baseline is critical for evaluating how much each controller improves comfort.

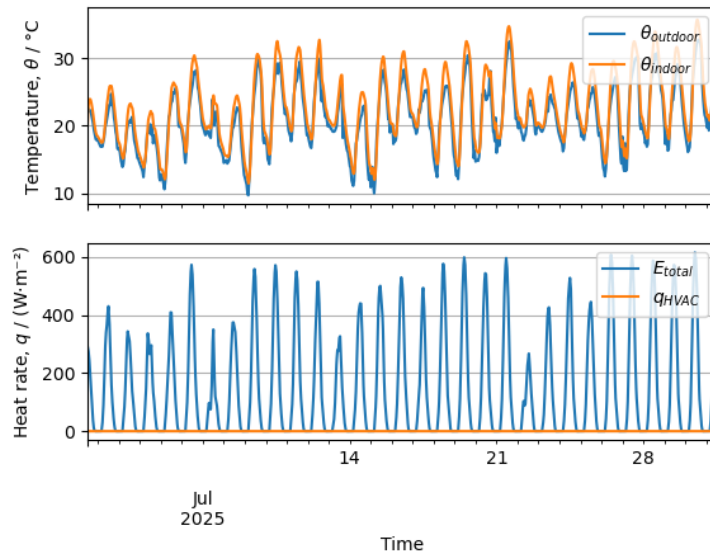


Figure 26: Free-running in the summer

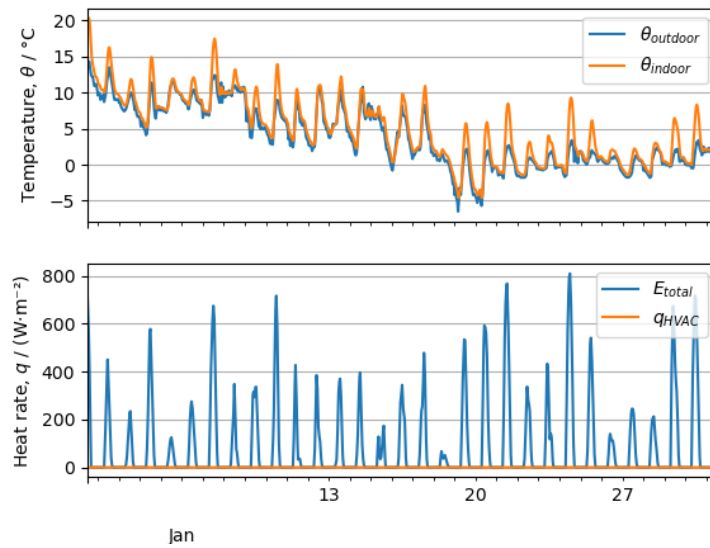


Figure 27: Free-running in the winter

10.1 Proportional nonlinear controller (PNC)

HEATING

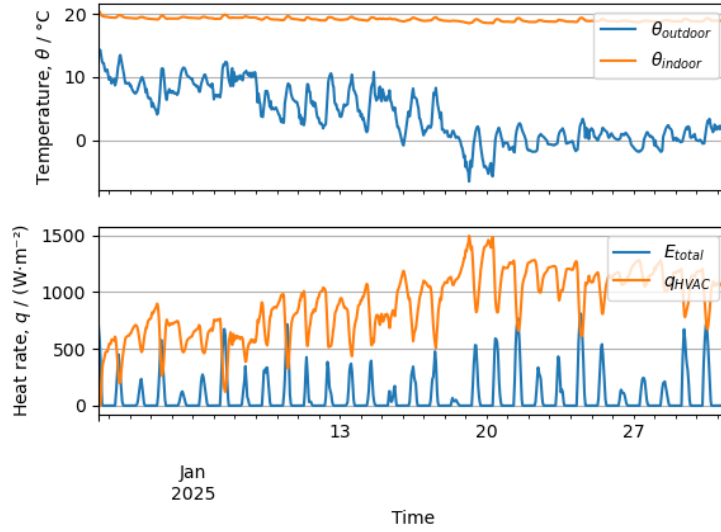


Figure 28: Heating in the winter - PNC

The PNC drives θ_{indoor} upward and holds it near the 20 $^{\circ}\text{C}$ setpoint, but the heat rate panel shows sustained high q_{HVAC} , peaking near $1500 \text{ W}\cdot\text{m}^{-2}$. Since the P-controller acts proportionally to the instantaneous error $e = \theta_{\text{set}} - \theta_{\text{indoor}}$, it applies strong corrective action when the outdoor-indoor temperature gap is large, regularly exceeding 20 $^{\circ}\text{C}$ in January. The indoor temperature tracks the setpoint closely but not exactly.

COOLING

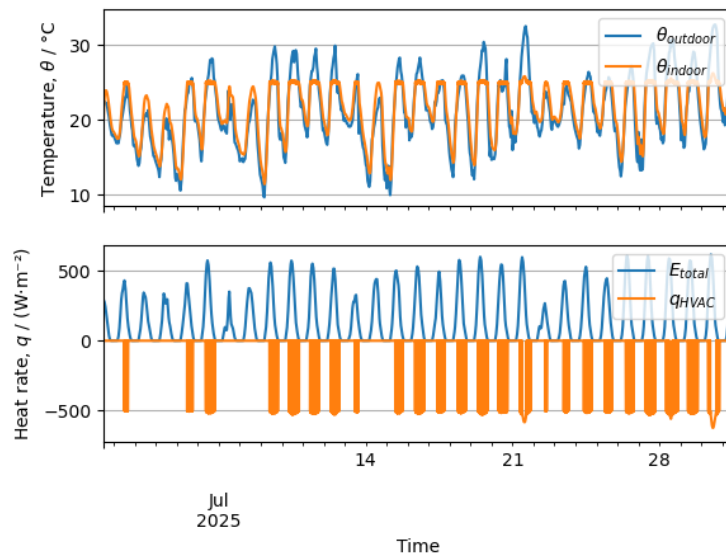


Figure 29: Cooling in the summer - PNC

The heat rate panel shows negative q_{HVAC} values (cooling mode) during the hottest daytime periods, with magnitudes up to approximately $-500 \text{ W} \cdot \text{m}^{-2}$. However, the controller reverts to zero during cooler nights, confirming that the PNC only reacts to the cooling needs.

HEATING AND COOLING

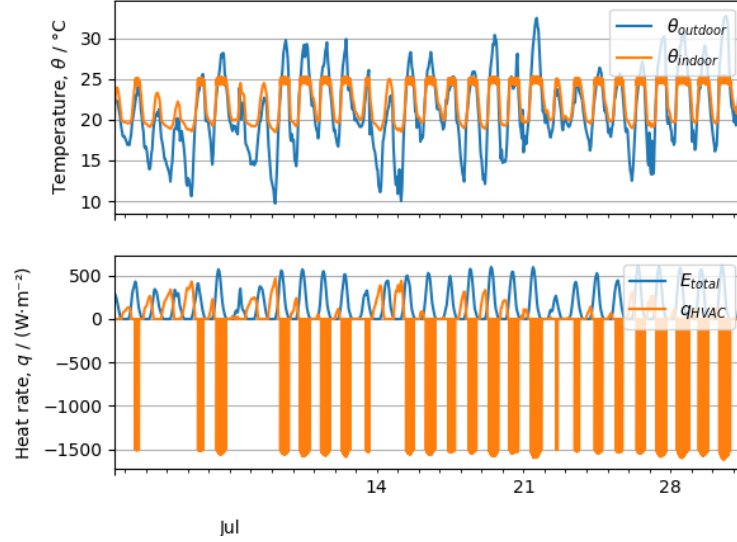


Figure 30: Heating and cooling in the summer - PNC

In summer, the PNC alternates between cooling and heating as outdoor temperature oscillates. The heat rate panel shows large bidirectional swings (-1500 to $+500 \text{ W} \cdot \text{m}^{-2}$), revealing that the PNC overshoots into heating territory after a cooling phase. This indicates chattering behaviour: the proportional-only law reacts to every disturbance without memory of past errors, causing unnecessary energy consumption.

SOLAR PROTECTION

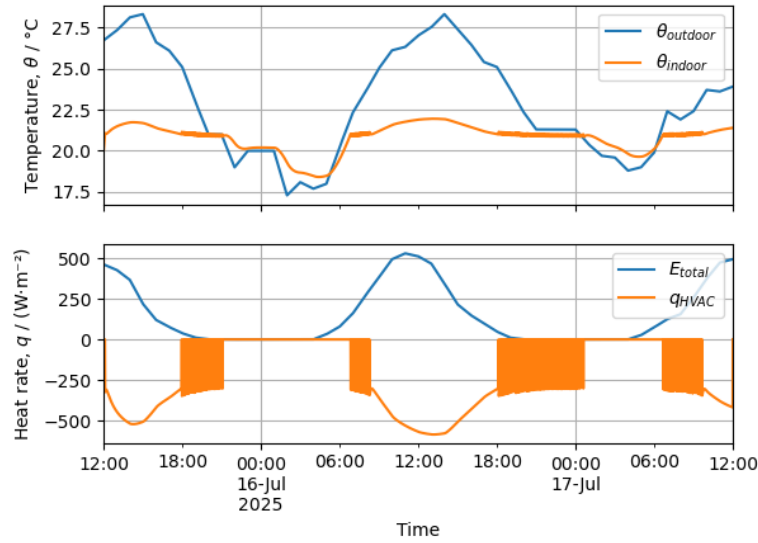


Figure 31: Solar protection - PNC

The PNC responds to the afternoon solar gain spike by increasing the cooling output, keeping θ_{indoor} below 27.5 °C. The heat rate panel shows clean transitions between cooling ($q_{HVAC} < 0$) and zero output during the cooler nighttime hours. While the setpoint is not perfectly maintained, the PNC prevents significant overheating during this high-irradiance event.

PASSIVE COOLING

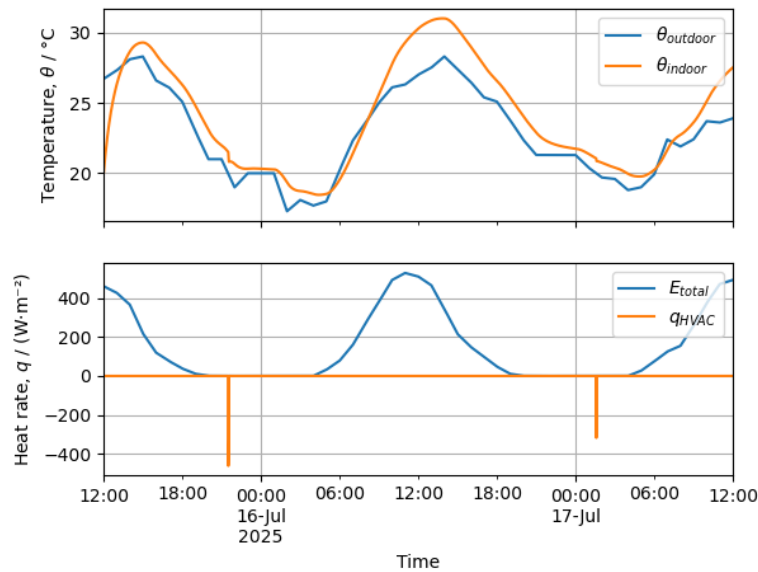


Figure 32: Passive cooling - PNC

During the passive cooling scenario, the PNC correctly deactivates the HVAC when natural ventilation is sufficient to maintain comfort, with q_{HVAC} remaining near zero for extended periods. The indoor temperature stays within approximately 20–27°C, and the controller only intervenes with moderate cooling pulses when the temperature drifts towards the upper bound. This confirms that the PNC can correctly exploit passive cooling opportunities, though its reactive nature leads to occasional unnecessary corrections.

10.2 Proportional-Integral with saturation nonlinear controller (PINC)

HEATING

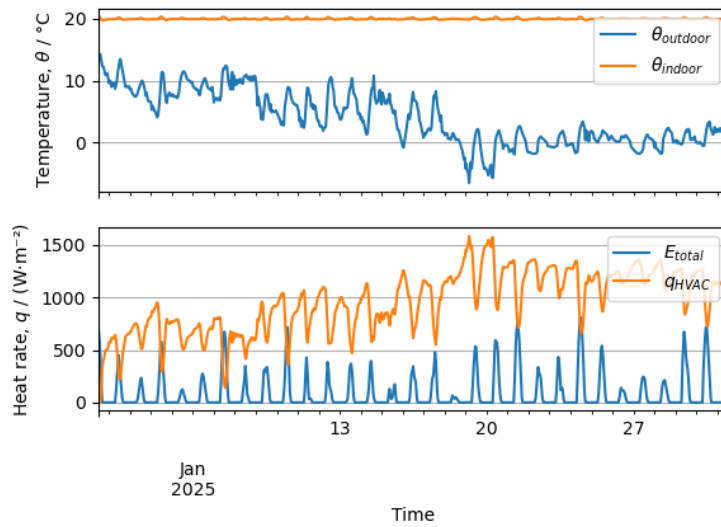


Figure 33: Heating in the winter - PINC

The PINC shows a very similar winter profile to the PNC, with indoor temperature stable near the setpoint and heat rate ranging from 0 to 1500 W·m⁻². The key difference is the integral term: it accumulates the error over time and eliminates any steady-state offset, so θ_{indoor} locks onto the setpoint exactly.

COOLING

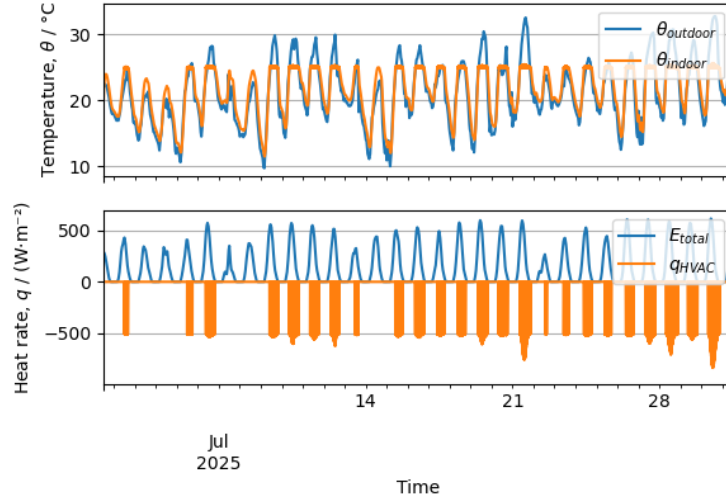


Figure 34: Cooling in the summer - PNC

Compared to the PNC, the PINC produces a smoother cooling response, with q_{HVAC} bounded within -500 to $+750 \text{ W} \cdot \text{m}^{-2}$. The integral action progressively builds up the corrective signal as the temperature error accumulates, avoiding the abrupt on/off switching seen in the PNC. The indoor temperature is maintained more consistently within the comfort band, reflecting the improved disturbance rejection brought by the PI structure.

HEATING AND COOLING

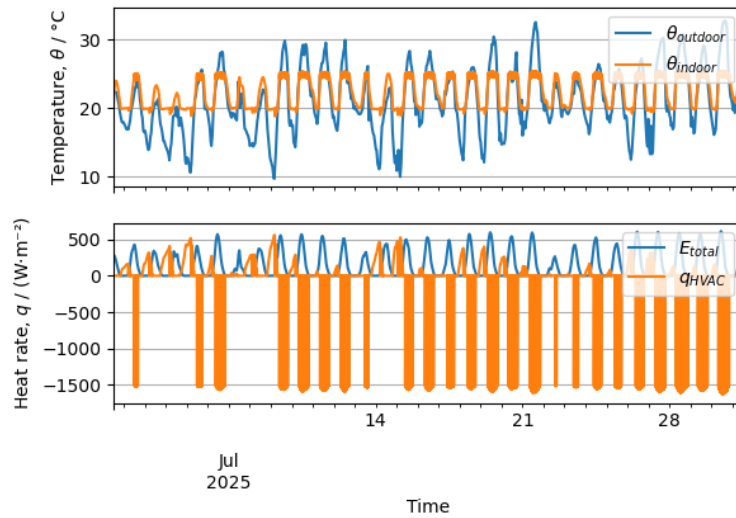


Figure 35: Heating and cooling in the summer - PNC

Compared to the PNC, the PINC heat rate in summer is smoother and better bounded, with bidirectional swings noticeably reduced. The integral action damps the oscillatory switching observed in the PNC by remembering the trend of the error, resulting in a more progressive and energy-efficient correction.

SOLAR PROTECTION

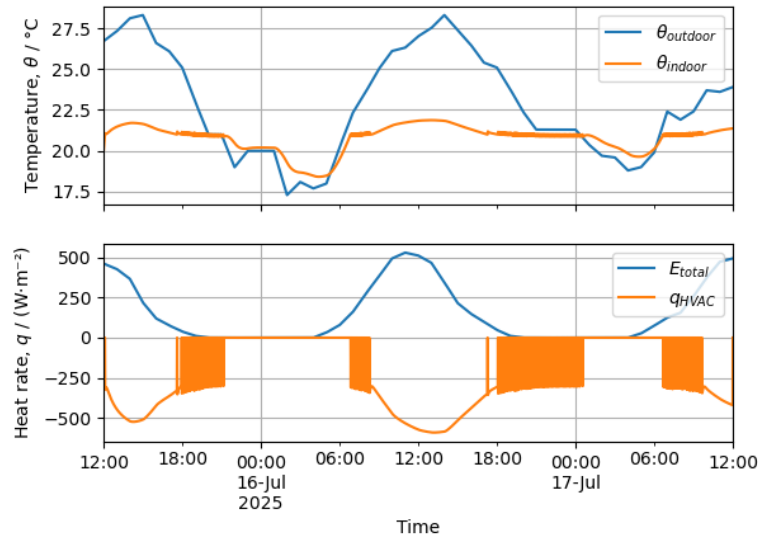


Figure 36: Solar protection - PINC

During the July solar protection scenario, the PINC contains the indoor temperature within a slightly narrower band than the PNC, with q_{HVAC} oscillating within approximately $-500 \text{ W}\cdot\text{m}^{-2}$. The smoother power profile indicates that the controller anticipates the sustained nature of the solar disturbance through its integral memory, modulating the cooling output progressively rather than reacting sharply to each instantaneous temperature spike.

PASSIVE COOLING

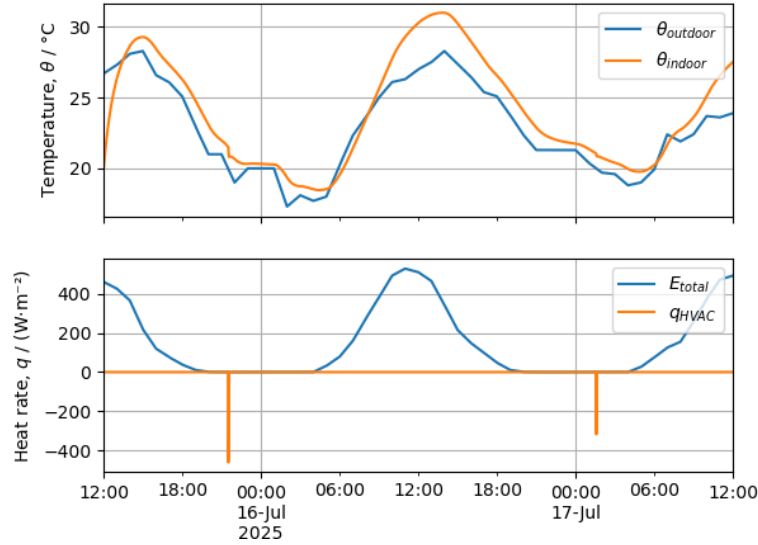


Figure 37: Passive cooling - PINC

In passive cooling mode, the PINC correctly clamps $q_{\text{HVAC}} \approx 0$ during periods of sufficient natural ventilation, thanks to the saturation mechanism that prevents the integrator from driving unnecessary heating when the system is near the comfort boundary.

11 Conclusion

The following key conclusions can be drawn from the complete thermal modelling, dynamic analysis, and control simulation of the 30 m² apartment in Grenoble, France.

Building Model and Thermal Parameters

- The overall thermal conductance is $H_{\text{total}} = 63.47 \text{ W/K}$, dominated by ventilation losses ($G_{12} = 30 \text{ W/K}$, representing 47% of the total), making it the single largest heat loss path.
- The window ($UA_{\text{win}} = 18.60 \text{ W/K}$, $U = 5.17 \text{ W/(m}^2 \cdot \text{K)}$) is the second largest loss path and the primary solar gain entry point, contributing up to 540 W of solar transmission at peak irradiance — 45.6% of the total summer cooling load.

Steady-State

- The peak winter heating load reaches $\dot{Q}_{\text{heat}} \approx 1904 \text{ W}$ and the peak summer cooling load reaches $\dot{Q}_{\text{cool}} \approx 1186 \text{ W}$, under the most unfavorable conditions, confirming the building is heavily heating-dominated.

Dynamic Analysis and Numerical Stability

- All five eigenvalues of the state matrix A_s are real and negative, confirming unconditional asymptotic stability of the uncontrolled system.
- The DAE and state-space models agree to a numerical error of $\sim 10^{-15}$ °C, validating the correctness of the matrix assembly and solver implementation.

Perfect Controller

- With $K_p = 10^4$ W/K, the perfect controller maintained θ_{indoor} within ± 0.1 °C of the setpoint in both seasons, switching cleanly between the 20 °C daytime setpoint and the 16 °C night setback.
- The high controller gain stiffened the system dynamics, reducing the maximum stable time step from 1800 s to just 20 s, illustrating the fundamental trade-off between control accuracy and simulation efficiency.

Nonlinear Controllers: PNC vs PINC

- Free-running simulations confirmed that passive thermal behaviour is entirely insufficient for comfort: without HVAC, indoor temperature reached 33.5 °C in summer and -2.9 °C in winter.
- The PNC maintained comfort in both seasons but produced a persistent steady-state offset and pronounced chattering in combined heating and cooling mode, with heat rate swings indicating unnecessary energy usage.
- The PINC eliminated the steady-state offset through integral action and prevented integrator windup via anti-windup, producing a smoother heat rate profile.
- The PINC is the recommended strategy for practical implementation, as it combines zero steady-state error, reduced energy waste, and robust behaviour during transient events such as solar gain spikes and passive cooling transitions.

Outlook and Recommendations

- Future work could explore other types of controllers, which leverages weather forecasts to pre-condition the building thermal mass, potentially reducing peak loads and further smoothing the HVAC power profile compared to the reactive PNC and PINC strategies evaluated in this report.

References

- [1] Ghiaus, C. (2024). dm4bem. https://cghiaus.github.io/dm4bem_book/tutorials.